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December 19, 2022

Dr. Christian Schilling,
Munich Center for Quantum Science and Technology,
Ludwig-Maximilians-Universität München.

Dear Dr. Christian Schilling,

I am writing to express my interest in joining your research group as a post-doctoral student. I am currently a PhD Candidate working under the supervision of Rob W. Spekkens in the Quantum Foundations group at the Perimeter Institute for Theoretical Physics in Waterloo, Ontario, Canada. I will be graduating April of 2023.

The majority of my research over the last few years has focused on the quantum marginal problem (and a few related topics) from the perspective of quantum information theory and quantum foundations. Consequently, many of the motivations for working on the quantum marginal problem (or more specifically, the n-representability problem) arising in quantum chemistry are new to me. Nevertheless, I am seeking opportunities to apply what I've learned to active topics in quantum chemistry and quantum many-body physics.

I look forward to discussing these motivations and connections with you further.

Included in this document are i) a research statement summarizing my past research accomplishments and future research plans, and ii) a copy of my CV which includes a publication list and contact details for two references.

Thank you for your time and consideration.

Sincerely,

Thomas C. Fraser

Research Statement

At the heart of many conceptual and technical challenges faced by quantum physicists is the quantum marginal problem (QMP). Generally speaking, the QMP aims to characterize the relationships between the various reduced states, or marginals, of a large multipartite or many-body quantum system. I believe further development of analytic and numerical tools which find the right balance between accuracy and efficiency are necessary to overcome these challenges.

The particular route through which I discovered and became interested in tackling the QMP, while rather atypical, perfectly summarizes my academic journey thus far.

Initially, I worked on problems from the field of causal modelling (Pearl 2009). One such problem is to derive necessary and sufficient conditions for a multivariate probability distribution to be generated by a graphical network of cause-effect influences between those variables. These conditions are typically formulated as polynomial inequalities among the probabilities assigned to elementary events. Importantly, these inequalities can sometimes differ depending on whether the causal-dependencies are described by classical physics or quantum physics, and demonstrating such a difference is an important prerequisite for establishing any quantum advantage in network scenarios. The prototypical example of such a difference is exhibited by the Bell-network and its associated Bell-inequality violations (Wood and Spekkens 2015). One of the leading strategies for deriving these inequalities, called the Inflation technique, is to first convert the problem into a family of classical (or quantum) marginal problems and then afterwards solve the resulting marginal problems (Wolfe, Spekkens, and Fritz 2019).

My first research project was to use this new technique on a particular network involving three variables, called the triangle-network, and to derive the analogue of Bell-inequalities for that network (Fraser and Wolfe 2018). Behind the scenes, I used a variety of computational techniques (for enumerating facets from a vertex description of a convex polytope) to solve the classical marginal problem. Unfortunately, when the number of random variables and their cardinalities become large, an exhaustive enumeration of all facets, while analytically correct, becomes computationally intractable even after accounting for numerous symmetries. In an effort to circumvent some of these obstacles, I stopped approaching the problem from the perspective of real algebraic geometry and instead developed a new approach for classical causal modelling that emphasized the combinatorial nature of the problem (Fraser 2020). This approach led to the resolution of a few open conjectures in the field and to the development of a convergent hierarchy of computationally tractable relaxations of the causal compatibility problem.

Later in 2020, I worked with D. Schmid et. al. on the foundations of entanglement theory (Schmid et al. 2020) from the resource-theoretic perspective (Coecke, Fritz,

and Spekkens 2016). In addition to emphasizing the importance of considering causal structure when defining and quantifying entanglement, we managed to show that the problem of deciding inter-conversion between multipartite pure states under local quantum operations could, at least partially, be reduced to the unipartite QMP.

As a result of these experiences, it became evident to me that the QMP was worth solving. In addition to the aforementioned applications of the QMP to quantum causal modelling and quantum entanglement, further applications to quantum chemistry, quantum many-body physics, and quantum error correction have motivated me further.

As you know, over the last two decades, tremendous progress has been made toward solving the QMP for the special case of disjoint subsystems (Klyachko 2006). These advances were made possible by deep connections between the representation theories of Lie groups and the convex geometry that emerges from the semi-group structure associated highest-weight irreducible representations. One manifestation of this connection, relating the spectra of quantum states to the isotypic components of Schur-Weyl decompositions, can be equipped with an operational significance through Keyl & Werner's work on spectral estimation (Keyl and Werner 2005). This correspondence was strengthened even further to make partial progress toward the overlapping QMP (Christandl, Şahinoğlu, and Walter 2018).

Following these developments, in my latest paper (Fraser 2022), I leveraged results for quantum state estimation (Keyl 2006) to derive a complete hierarchy of inequality constraints which characterize the set of compatible quantum marginals.

Given my previous research experiences, I would be best suited working on either the long-term research project P3 or P4, although I'd be eager to learn more about all four research projects. In the future, there are a few additional topics/questions I hope to work on:

- The compatibility inequalities in (Fraser 2022) are efficiently computable (at least for low degree). When restricted to fermionic or bosonic multipartite pure states and unipartite marginals, how do these conditions compare with existing techniques?
- Which quantum states exhibit overlapping marginals that live on the boundary of the compatible region and what is their physical or information-theoretic significance?
- For disjoint marginals, there exists a geometric interpretation of the inequality constraints as the intersection of certain Schubert varieties. What is the corresponding geometric interpretation for overlapping marginals?
- How generic is the correspondence between tensor network models and maximum entropy states compatible with given marginals (Kim 2021)?

References

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